

BUILDINGS OF GENERALIZED BRAID GROUPS

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(1.24)–(3.6)

(1.24) Proposition. Let A be a chamber, let P be an intersection of walls of A , let $F = F(P)$ (1.6), and let g be an equivalence class of galleries with source A . There exists a gallery class g' of $(V_P, \mathcal{M}_P)$, with source $\pi_F^{-1}(A)$, such that, for every gallery H of $(V_P, \mathcal{M}_P)$, g begins with $\pi_F(H)$ if and only if g' begins with H .

Let $\mathcal{S}$ be the set of gallery classes h with source $\pi_F^{-1}(A)$ in $(V_P, \mathcal{M}_P)$ such that g begins with $\pi_F(h)$. Apply (1.14) to $(V_P, \mathcal{M}_P)$ and to $\mathcal{S}$. By (1.19), condition (i) is satisfied, cf. the proof of (1.20), and (1.24) follows from (1.14)(ii).

(1.25) The set of galleries may be regarded as the set of arrows of a category $\text{Gal}_0(V, \mathcal{M})$ whose objects are the chambers. Similarly, equivalence classes of galleries are the arrows of a quotient category $\text{Gal}^+(V, \mathcal{M})$. Let A, B, C be three chambers, let E be a gallery from A to B , and let F be a gallery from B to C . Although the general categorical convention is different, we shall continue to write EF for the composite of E and F . The law $G \mapsto G^*$, respectively $G \mapsto -G$, is an anti-equivalence, respectively an equivalence, of $\text{Gal}_0(V, \mathcal{M})$, or of $\text{Gal}^+(V, \mathcal{M})$, with itself, inducing the identity, respectively $C \mapsto -C$, on objects.

When no confusion is possible, we shall write simply Gal_0 and Gal^+ for $\text{Gal}_0(V, \mathcal{M})$ and $\text{Gal}^+(V, \mathcal{M})$, or for a category $\text{Gal}_0(V_P, \mathcal{M}_P)$ or $\text{Gal}^+(V_P, \mathcal{M}_P)$. By abuse of language, we shall often call the arrows of Gal^+ galleries.

(1.26) Lemma.

(i) In Gal^+ , for every gallery g , one has

$$g\Delta = \Delta(-g).$$

(ii) For any g and h with source A in Gal^+ , there exists n such that $g\Delta^n$ begins with h . If h is composed of k galleries $u(A_i, A_{i+1})$, one may take $n = k$.

By induction, it is enough to prove (i) when g has length one. For $g = (B, C)$, one has

$$g\Delta = u(B, C)u(C, -C) = u(B, C)u(C, -B)u(-B, -C) = u(B, -B)u(-B, -C) = \Delta(-g).$$

For every chamber B , $\Delta = u(A, -A)$ begins with $u(A, B)$. Hence (ii) follows from (i) by induction on k .

The following proposition follows at once from (1.26), from its transform by $*$, and from (1.19)(i),(ii).

(1.27) Proposition.

(i) In Gal^+ , the set of all arrows admits a left and right calculus of fractions.

(ii) Let $\text{Gal}(V, \mathcal{M})$, or simply Gal , be the category, a groupoid, obtained from Gal^+ by making all arrows invertible. Call positive those arrows of Gal lying in the image of Gal^+ . The canonical functor from Gal^+ to Gal is faithful, and every arrow g of Gal can be written in the forms

$$g = g_1\Delta^{-n} = \Delta^{-n}g_2$$

with g_1 and g_2 positive.

If a category C has only one object A , and if the monoid $\text{Hom}(A, A)$ is cancellative, to say that all arrows of C admit a left and right calculus of fractions means that $\text{Hom}(A, A)$ satisfies Ore's condition on the left and on the right. The calculus of fractions used in (1.27) and below is an immediate generalization of Ore's theory for embedding such a monoid in a group.

(1.28) For every wall M , the number of times that $g \in \text{Gal}^+$ crosses M is well-defined (1.11). This function of g extends additively to $g \in \text{Gal}$. More generally, for every intersection of walls P , one could use the universal property of Gal to define functors

$$\pi_P : \text{Gal}(V, \mathcal{M}) \longrightarrow \text{Gal}(V_P, \mathcal{M}_P).$$

(1.29) Let A be a chamber, let P be an intersection of walls of A , and let $F = F(P)$ (1.6). The function π_F induces a functor

$$\pi_F : \text{Gal}_0(V_P, \mathcal{M}_P) \longrightarrow \text{Gal}_0(V, \mathcal{M}). \quad (1.29.1)$$

The image of this functor is stable under equivalence, and it induces a faithful functor

$$\pi_F : \text{Gal}^+(V_P, \mathcal{M}_P) \longrightarrow \text{Gal}^+(V, \mathcal{M}). \quad (1.29.2)$$

From the calculus of fractions and (1.19)(i) follows the following proposition.

(1.30) Proposition. Under the preceding hypotheses, the functor deduced from (1.29.2)

$$\pi_F : \text{Gal}(V_P, \mathcal{M}_P) \longrightarrow \text{Gal}(V, \mathcal{M})$$

is faithful.

(1.31) Proposition. Let C be a chamber; let I and J be two sets of walls of C ; put $K = I \cup J$; let P, Q, R be the intersections of the walls in I, J, K , and let $F(P), F(Q), F(R)$ be the corresponding facets in C (1.6). The facet $F(R)$ is the smallest facet containing $F(P)$ and $F(Q)$. In the groupoid $\text{Gal}(V, \mathcal{M})$, one has

$$\pi_{F(P)}(\text{Gal}(V_P, \mathcal{M}_P)) \cap \pi_{F(Q)}(\text{Gal}(V_Q, \mathcal{M}_Q)) = \pi_{F(R)}(\text{Gal}(V_R, \mathcal{M}_R)).$$

A chamber having $F(P)$ and $F(Q)$ as facets also has $F(R)$ as a facet. This proves (1.31) for objects.

Let $g \in \text{Hom}_{\text{Gal}}(A, B)$, and suppose that

$$g = \pi_{F(P)}(g_1) = \pi_{F(Q)}(g_2).$$

By (1.27)(ii), one has

$$g_1 = \Delta(P)^{-n} g'_1, \quad g_2 = \Delta(Q)^{-n} g'_2,$$

with g'_i positive, for $n > 0$ sufficiently large. Let $g_1 = \pi_{F(P)}(g'_1)$ and $g_2 = \pi_{F(Q)}(g'_2)$. By (1.26)(ii), transformed by $*$, $\Delta^n \Delta(P)^{-n}$ is positive, and in $\text{Gal}^+(V, \mathcal{M})$ one has

$$(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2. \quad (1.31.1)$$

(1.31.2) Lemma. If $h \in \text{Hom}_{\text{Gal}^+}(A, B)$ begins both with $\Delta^n \Delta(P)^{-n}$ and with $\Delta^n \Delta(Q)^{-n}$, then h begins with $\Delta^n \Delta(R)^{-n}$.

We deduce (1.31) from (1.31.2). By (1.31.2), one has

$$(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2 = (\Delta^n \Delta(R)^{-n}) g_3$$

with g_3 positive. Hence $g = \Delta(R)^{-n} g_3$. The positive gallery $g_3 = \Delta(R)^n g$ lies in the image of both $\pi_{F(P)}$ and $\pi_{F(Q)}$. By (1.28), it therefore crosses zero times every wall not containing P or Q , i.e. every wall not containing R . It thus lies in the image of $\pi_{F(R)}$.

We prove (1.31.2). Let $A(P) = A \cdot \Delta \cdot A(P)$, and similarly for Q and R . The gallery $\Delta \Delta(P)^{-1}$ with source A is $u(A, A(P))$; the one with source $A(P)$ is $u(A(P), A)$. By (1.26)(i), $\Delta \Delta(P) = \Delta(P) \Delta$. Hence

$$\Delta^n \Delta(P)^{-n} = u(A, A(P)) u(A(P), A) u(A, A(P)) \cdots \quad (1.31.3)$$

with n factors.

(1.31.4) Lemma. $A(R) \in D(A(P), A(Q))$.

With the notation of (1.2), $\mathcal{M}(A, A(P))$ is the set of walls containing P , and $\mathcal{M}(A, A(Q))$ is the set of those containing Q . By (1.2), $\mathcal{M}(A(P), A(Q))$ is the set of walls containing P or Q , but not R , namely

$$\mathcal{M}(A(P), A(R)) \cup \mathcal{M}(A(Q), A(R)),$$

and one applies (1.4).

By (1.19)(iii), a gallery class which begins with $\Delta \Delta(P)^{-1}$ and $\Delta \Delta(Q)^{-1}$ also begins with $\Delta \Delta(R)^{-1}$, and (1.31.2) follows by induction from the following lemma.

(1.31.5) Lemma. Let P and R be intersections of walls of a chamber A , with $P \subset R$. Let h be in Gal^+ with source A . If h begins both with $\Delta^n \Delta(P)^{-n}$, $n > 0$, and with $\Delta \Delta(R)^{-1}$, then h begins with

$$(\Delta \Delta(R)^{-1})(\Delta \Delta(P)^{-1})^{n-1}.$$

One may assume $n > 2$. Put $h = (\Delta \Delta(P)^{-1})h'$. With the preceding notation, h' then begins both with $\Delta \Delta(P)^{-1} = u(A(P), A)$ and with $u(A(P), A(R))$. Apply (1.19)(iii). The chamber C of loc. cit. is then separated from $A(P)$ by every wall M separating $A(P)$ from A , i.e. $M \not\supset P$, or $A(P)$ from $A(R)$, i.e. $M \supset P$ and $M \not\supset R$. Thus C is separated from $A(P)A$ only by walls $M \supset R$, and

$$C \in D(A(P)A, A(P)A(R)).$$

It follows that h' begins both with $\Delta \Delta(R)^{-1}$ and with $(\Delta \Delta(P)^{-1})^{n-1}$. Induction on n shows that h' begins with

$$(\Delta \Delta(R)^{-1})(\Delta \Delta(P)^{-1})^{n-2},$$

so that h begins with

$$(\Delta \Delta(P)^{-1})(\Delta \Delta(R)^{-1})(\Delta \Delta(P)^{-1})^{n-2}.$$

One concludes by noting that

$$\Delta \Delta(P)^{-1} \Delta \Delta(R)^{-1} = \Delta \Delta(R)^{-1} \Delta \Delta(P)^{-1}.$$

(1.32) Proposition. For F a facet of a chamber C , generating an intersection of walls P , one has

$$\pi_F^{-1}(\text{Gal}^+(V, \mathcal{M})) = \text{Gal}^+(V_P, \mathcal{M}_P) \subset \text{Gal}(V_P, \mathcal{M}_P).$$

Suppose that $\pi_F(\Delta^{-n}g) = h$ lies in $\text{Gal}^+(V, \mathcal{M})$. Then $\pi_F(g) = \Delta(P)^n h$, and by (1.28) every gallery H representing h crosses only walls passing through F . Thus $h = \pi_F(h_1)$, with h_1 in $\text{Gal}^+(V_P, \mathcal{M}_P)$, and the conclusion follows from (1.30).

2. Buildings

(2.1) Let $(V, \mathcal{M})$ be as in §1, satisfying (1.8), and let $r = \dim V$. Let S be the sphere of radius one in V , for some Euclidean structure on V ; more intrinsically, one could take $S = (V - \{0\})/\mathbb{R}_+^*$. The hyperplanes $M \in \mathcal{M}$ cut out a triangulation of S , and we shall again denote by S both the corresponding simplicial space and its geometric realization. We transfer to S the terminology used for V : chambers, facets, adjacency, galleries.

(2.2) Choose a fundamental chamber A_0 in S . We denote by I^+ the space constructed below; it depends on $V, \mathcal{M}, A_0$, and is equipped with $q : I^+ \rightarrow S$.

a) Let $\mathcal{G}^+$ be the set of equivalence classes of galleries g with source A_0 ,

$$\mathcal{G}^+ = \coprod_B \text{Hom}_{\text{Gal}^+}(A_0, B).$$

b) Let Z^+ be the disjoint sum, indexed by $g \in \mathcal{G}^+$, of the closed simplex of S which is the closure of the open simplex of dimension $r - 1$ equal to the target of g :

$$Z^+ = \coprod_{g \in \mathcal{G}^+} (\text{target } g)^-.$$

Let q' be the evident map from Z^+ to S .

c) I^+ , with q , is a quotient of Z^+ , with q' . If G is a gallery from A_0 to B , and if C is a chamber adjacent to B , one glues $(\text{target } g)^-$ and $(\text{target } g(BC))^-$ along the common closed face of their images in S .

(2.3) The space I^+ is decomposed into facets, images of facets of Z^+ , each mapping bijectively to a facet of S . Its chambers, respectively faces and vertices, are its facets of dimension $r - 1$, respectively $r - 2$ and 0 .

The vertices of a facet F are the vertices contained in the closure $\overline{F}$ of F . The chambers of I^+ are indexed by $\mathcal{G}^+$; $A_0.g$ denotes the one indexed by g , and A_0 the one indexed by (A_0) .

Let $g \in \text{Hom}_{\text{Gal}^+}(A_0, A)$, let $A = A_0.g$, and let $H = (C_0, \dots, C_n)$ be a gallery from A to B , with class h . Put

$$A.h = A_0.gh.$$

The chambers $A.(C_0, \dots, C_i)$, $0 \leq i \leq n$, form a gallery in I^+ . Such galleries will be called positive.

(2.4) Lemma. Let F be a facet of S , let A and B be two chambers of S having F as a facet, and let $g \in \text{Hom}_{\text{Gal}^+}(A_0, A)$, $h \in \text{Hom}_{\text{Gal}^+}(A_0, B)$. The following conditions are equivalent:

- (a) in I^+ , the facets $q^{-1}(F)$ of $A_0.g$ and $A_0.h$ coincide;
- (b) $g^{-1}h \in \text{Hom}_{\text{Gal}}(A, B)$ lies in $\pi_F(\text{Gal})$;
- (c) there are g_1 and h_1 in $\pi_F(\text{Gal}^+)$ such that $gg_1 = hh_1$.

Condition (b) is an equivalence relation between galleries from A_0 to a chamber having F as a facet. This gives (a) $\Rightarrow$ (b). That (b) implies (c) follows from (1.26)(i). Finally, (c) $\Rightarrow$ (a) is immediate from the definitions.

From (2.4), (a) $\Leftrightarrow$ (b), and from (1.31), one immediately obtains:

(2.5) Proposition. Each facet of I^+ is uniquely determined by the set of its vertices. Thus I^+ may be described as the geometric realization of the following simplicial scheme.

- a) For x a vertex of S , let $\mathcal{G}(x)$ be the set of g in Gal^+ with source A_0 and with target a chamber having x as a vertex. Let $\mathcal{G}(x)/\pi_x$ be the quotient of $\mathcal{G}(x)$ by the equivalence relation (2.4)(b), for $F = x$. Then the set of vertices is

$$\coprod_x \mathcal{G}(x)/\pi_x.$$

- b) A set E of vertices spans a simplex if and only if there exists a gallery g with source A_0 such that, for $y \in E$, g lies in $\mathcal{G}(qy)$ and y is the image of g in $\mathcal{G}(qy)/\pi_{qy}$.

(2.6) Proposition. Let F be a facet of I^+ . There exists a gallery class g such that, for F to be a facet of a chamber $B = A_0.h$, it is necessary and sufficient that

$$h = g\pi_F(h')$$

for a suitable h' in Gal^+ .

Take g to be a gallery of minimum length such that F is a facet of $A = A_0.g$. Let $B = A_0.h$ be a chamber having F as a facet. By (2.4)(a) $\Leftrightarrow$ (c), there are galleries h_1 and h_2 in $(V_{qF}, \mathcal{M}_{qF})$ with

$$g\pi_F(h_1) = h\pi_F(h_2).$$

Apply the transform by $*$ of (1.24). By the minimality of g , one finds that h_1 ends with h_2 , say $h_1 = h'h_2$. By (1.19)(ii), $g\pi_F(h') = h$, which proves (2.6).

(2.7) Notation. If A is a chamber of I^+ , denote by $S(A)$ the union of the closed chambers $(A.(qA, B))^-$, where B runs through the chambers of S . One checks that $q|S(A)$ is an isomorphism from $S(A)$ to S .

(2.8) Definition. $\bar{I}^+$ is the space obtained from I^+ by filling in the spheres $S(A)$.

More precisely, let B^r be a ball whose boundary is S . If one wants to work simplicially, one takes B^r to be the cone on the simplicial space S . Define $\bar{I}^+$ from I^+ by attaching a family of copies $b(A)$ of B^r , indexed by the chambers of I^+ .

(2.9.2) Lemma. Suppose that there exists a chamber $B = A_0.h$ in $(I^+)_{n+1}$, distinct from $A = A_0.g$, having F as a facet. Then there exists a chamber $B' = A_0.h'$ in $(I^+)_n$, and a wall M of qB' , containing qF , such that

$$A = B'.\Delta(M).$$

Let g_0 be the gallery considered in (2.6). One has

$$g = g_0\pi_F(g_1), \quad h = g_0\pi_F(h_1).$$

The hypotheses imply that $\lg(g_1) \neq 0$. For a suitable wall M of qA , one then has $g_1 = h'\Delta(M)$, and one puts $B' = \pi_F h'$.

For the closed chamber $\bar{A}$, $\bar{A} \cap (I^+)_n$ is the union of a nonempty set of closed faces, and in $(I^+)_{n+1}$ the points of

$$\bar{A} - (\bar{A} \cap (I^+)_n)$$

belong only to the single closed chamber $\bar{A}$. There are two cases.

Case 1. $\bar{A} \cap (I^+)_n \neq \partial\bar{A}$. In this case, there is no sphere $S(B)$ with

$$A \subset S(B) \subset (I^+)_{n+1}.$$

We take $\varphi_t|\bar{A}$ to be a collapse of $\bar{A}$ onto $\bar{A} \cap (I^+)_n$.

Case 2. $\bar{A} \cap (I^+)_n = \partial\bar{A}$. Put $A = A_0.g$. For every wall M of qA , it follows from (2.7.2) that g ends with $\Delta(M)$. By (1.19)(iii), g therefore ends with Δ : $A = B.\Delta$, with B uniquely determined by A , by (1.19)(ii). In passing from $(\bar{I}^+)_{n+1}$ to $(\bar{I}^+)_n$, A and the interior of the ball $b(B)$ disappear. We take $\varphi_t|b(B)$ to be a collapse of $b(B)$ onto $S(B) - A$. As seen in case 1, every ball which disappears is of the preceding type. This completes the construction of φ_t , and proves (2.7).

(2.10) The space I , with $q : I \rightarrow S$, is defined like I^+ , replacing Gal^+ by Gal :

- a) put $\mathcal{G} = \coprod_B \text{Hom}_{\text{Gal}}(A_0, B)$;
- b) put $Z = \coprod_{g \in \mathcal{G}} (\text{target } g)^-$;
- c) let q' be the evident map from Z to S . The space I , with q , is a quotient of Z , with q' . If $g \in \mathcal{G}$ has target B and C is adjacent to B , one glues $(\text{target } g)^-$ and $(\text{target } g(BC))^-$ along the common closed face of their images in S .

Equivalently, if two chambers B and C have a common facet F , if $g \in \text{Hom}_{\text{Gal}}(A_0, B)$, and if $h \in \text{Hom}_{\text{Gal}}(A_0, C)$, then, in I , the closed facets $q'^{-1}(F)^-$ of $(\text{target } g)^-$ and of $(\text{target } h)^-$ are identified if and only if

$$hg^{-1} \in \pi_F \text{Hom}_{\text{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C).$$

(2.11) As for I^+ , one defines chambers, faces, and open or closed facets of I . One defines A_0 , and $A.h$, for A a chamber and h in Gal with source qA , as in (2.3). The analogue of (2.4)(a) $\Leftrightarrow$ (b) is here trivially true. As in (2.5), one deduces from (1.31) that each facet of I is uniquely determined by its set of vertices, and I appears as the geometric realization of a simplicial scheme with a description of type (2.5). For every chamber A of I , define a sphere $S(A)$ as in (2.7); q induces an isomorphism from $S(A)$ to S . As in (2.8), define a building $\bar{I}$, equipped with $q : \bar{I} \rightarrow B^r$, by filling in all the spheres $S(A)$.

(2.12) Besides q and its chamber decomposition, I has the following additional structure. A composition $B.g$ is defined, which associates to a chamber B of I , whose image in S is B , and to $g \in \text{Hom}_{\text{Gal}}(B, C)$, a chamber of I whose image in S is C . The construction $g \mapsto B.g$ identifies the analogue of I obtained by choosing B as fundamental chamber with the space I . In particular,

$$B.(gh) = (B.g).h.$$

(2.13) For every chamber A of I above A_0 , define a map

$$i_A : I^+ \longrightarrow I$$

and similarly $i_A : \bar{I}^+ \rightarrow \bar{I}$, compatible with the projection to S , respectively B^r , by sending the closed chamber $(A.g)^-$ of I^+ , respectively the ball $b(A.g)$ of $\bar{I}^+$, to the chamber, respectively ball, of the same name in I , respectively $\bar{I}$.

(2.14) Proposition.

- (i) i_A identifies I^+ with a closed subspace of I , and $\bar{I}^+$ with a closed subspace of $\bar{I}$.
- (ii) One has

$$I = \varinjlim i_{A_0, \Delta^{-2n}}(I^+), \quad \bar{I} = \varinjlim i_{A_0, \Delta^{-2n}}(\bar{I}^+).$$

The assertions (i) or (ii) for $\bar{I}^+$ and $\bar{I}$ follow from the corresponding assertions for I^+ and I .

We prove (i) for I^+ and I . Let B and C be two chambers of S with a common facet F . Let $g \in \text{Hom}_{\text{Gal}^+}(A_0, B)$ and $h \in \text{Hom}_{\text{Gal}^+}(A, C)$. Suppose that the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are equal in I . Then

$$g^{-1}h = \pi_F(e) \quad \text{for } e \in \text{Hom}_{\text{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C).$$

By (1.26), write $e = e_1 e_2^{-1}$, with e_i in Gal^+ . By (1.29), $g\pi_F(e_1) = h\pi_F(e_2)$ in Gal^+ , and the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are therefore already equal in I^+ .

For (ii), it is enough to note that every gallery g with source A_0 in Gal can be written

$$g = \Delta^{-2n} g'$$

with g' in Gal^+ .

From (2.14) one obtains $\pi_i(I) = \varinjlim \pi_i(I^+)$. Since I is a CW-complex, (2.9) gives:

(2.15) Theorem. The space $\bar{I}$ is contractible.

The space I is therefore the bouquet of the spheres $S(A)$, as A ranges through the chambers of I .

3. Coverings

(3.1) Let $V_{\mathbb{C}}$ be the complexification of V , and let $M_{\mathbb{C}}$ be the complexification of M , for $M \in \mathcal{M}$. Put

$$Y = V_{\mathbb{C}} - \bigcup_{M \in \mathcal{M}} M_{\mathbb{C}}.$$

In this section, we construct by gluing a space above Y . The gluing data will be described using I and $\bar{I}$. The facets of S and the facets of V other than $\{0\}$ are in canonical bijection, and we shall constantly pass from one to the other. In general we shall use the same notation for the corresponding facets; when necessary the correspondence will be denoted $F \mapsto [F]$.

(3.2) Lemma. Let A, B, C be three chambers of I , with $C \subset S(A) \cap S(B)$. Then q induces an isomorphism from $S(A) \cap S(B)$ to the intersection of S with the closed half-spaces $D'_M(qC)$, containing qC , bounded by a wall M which separates qA from qB .

The intersection of the $D'_M(qC)$ is a union of closed chambers. If C' is one of them, then qA and qB lie on the same side of every wall separating qC from C' . It follows that the lifts, to $S(A)$ or $S(B)$, of a minimal gallery from qC to C' coincide, and $(C')^- \subset q(S(A) \cap S(B))$.

Conversely, if a facet F is not in the intersection of the $D'_M(qC)$, then there is a wall M such that $F \subset M$, separating qA from qB and F from qC . Suppose, for contradiction, that $F \subset q(S(A) \cap S(B))$. Let D be a chamber of S having F as a facet, and let $(C_0, C_1, \dots, C_{n+1})$ be a gallery from qC to D . Let M_i be the wall separating C_i from C_{i+1} , and let α_i , respectively β_i , be $+1$ or -1 according as C_{i+1} and A , respectively B , are or are not on the same side of M_i . By hypothesis, F is a facet of

$$C.\Delta(M_0)^{\alpha_0} \dots \Delta(M_n)^{\alpha_n}$$

and of

$$C.\Delta(M_0)^{\beta_0} \dots \Delta(M_n)^{\beta_n}.$$

By (2.4) and (1.24), since $M \not\supset F$, this gives contradictory equalities: one side equals 1, the other -1 .

(3.3) Lemma. Let $\Re$ and $\Im$ be the real and imaginary part maps from $V_{\mathbb{C}}$ to V . Let $v \in V_{\mathbb{C}}$, and let F be the facet of V such that $\Re v \in F$. For v to belong to Y , it is necessary and sufficient that $\text{pr}_F(\Im v)$ lie in a chamber of $(V_F, \mathcal{M}_F)$.

This is clear.

What follows is based on the following intuition: to every walk in I , performed by following a gallery of the form

$$\Delta(M_1)^{\varepsilon_1} \cdots \Delta(M_k)^{\varepsilon_k}, \quad \varepsilon_i = \pm 1,$$

there corresponds a path, or path class, in Y . Its image under $\Re$ in V passes from chamber to chamber, crossing successively the walls $M_1, \dots, M_k$ at points m_i . For $\varepsilon_i = 1$, respectively -1 , the path passes through one of the two components R of $\Re^{-1}(m_i) \cap Y$, in bijection by $\Im$ with $V - M_i$: the one such that $\Im R$ contains, respectively does not contain, the preceding chamber.

(3.4) Notation.

- (i) If C is a chamber of $(V, \mathcal{M})$, let $Y(C)$ be the open subset of Y consisting of the $x \in V_{\mathbb{C}}$ satisfying the following condition: if F is the facet of V such that $\Re x \in F$, then $\text{pr}_F(\Im x) \in \pi_F^{-1}(C)$.
- (ii) Let A and B be two chambers of I . If $S(A) \cap S(B)$ contains a chamber C , temporarily denote by $Y'(A, B)$ the intersection of the open half-spaces $D'_M(qC)^{\circ}$ for M as in 3.2. Put

$$Y(A, B) = Y(qA) \cap Y(qB) \cap \Re^{-1}(Y'(A, B)) \subset V_{\mathbb{C}}.$$

If $S(A) \cap S(B)$ contains no chamber, put $Y(A, B) = \emptyset$.

- (iii) For a facet F of V , the star $\text{Et}(F)$ is the union of the open facets G of $(V, \mathcal{M})$ such that $F \subset \overline{G}$.
- (iv) For a facet F of V and a chamber C of $(V_F, \mathcal{M}_F)$, put

$$V(F, C) = \{v \in V_{\mathbb{C}} \mid \Re v \in \text{Et}(F), \text{pr}_F(\Im v) \in C\}.$$

(3.5) Lemma. Let G be a facet of V . The $V(F, C)$, for F a facet such that $G \subset \overline{F}$ and C a chamber of V_F , form an open covering of

$$Y \cap \Re^{-1}(\text{Et } G).$$

In particular, taking $G = \{0\}$, the $V(F, C)$ cover Y .

If $x \in Y \cap \Re^{-1}(\text{Et } G)$ and $\Re x \in F$, then x lies in one of the $V(F, C)$, by (3.3).

(3.6) Let Y' be the disjoint sum, indexed by the balls $b(A)$ of $\overline{I}$,

$$Y' = \coprod_{b(A)} Y(qA).$$